

ReflectAI: Promoting Student Wellbeing using LLM-Driven Multimodal Sensing System for Proactive Behavioral Feedback

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Abstract

University students today face a growing mental health crisis, yet existing wellbeing technologies often operate in isolation, rely heavily on manual self-tracking, and struggle to detect subtle behavioral changes or provide timely, personalized support. To address these limitations, we present ReflectAI, a proactive digital companion that combines multimodal passive sensing with large language model (LLM) reasoning to deliver context-aware, empathetic wellbeing feedback by integrating two key components: a contextual recognizer that semantically interprets behavioral cues from sensor and schedule data, and an LLM-based reasoning engine that infers user intent and generates psychologically informed responses. Leveraging few-shot prompting and pretrained psychological knowledge, the system interprets behavioral summaries—such as reduced mobility or irregular sleep—and delivers timely, adaptive feedback without requiring manual input or rule-based classifiers. Preliminary evaluation on StudentLife and GLOBEM datasets demonstrate promising output quality, validating the feasibility of LLM-powered behavior-aware feedback generation.

CCS Concepts

• Human-centered computing; • Applied computing → Health care information systems;

Keywords

Mental health, Passive sensing, LLMs, Personalization, Wellbeing

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1 Introduction

University students globally are experiencing a worsening mental health crisis, with rising rates of stress, burnout, and social withdrawal that undermine academic performance and wellbeing [4]. Early and personalized mental health support is critical to address these challenges. To this end, mobile health applications have become increasingly popular, offering tools for symptom tracking, activity logging, and self-monitoring [5]. However, these applications largely require active user engagement, such as self-reporting mood or manually entering data. This dependence on active participation renders them reactive and burdensome, particularly during periods of emotional fatigue or high academic pressure, when students are least likely to engage. As a result, adherence is low, and the opportunity to deliver timely interventions is often missed.

Meanwhile, large language models (LLMs) [1, 8] such as have demonstrated remarkable capabilities in natural language understanding, contextual reasoning, and generating human-like responses across diverse domains, including healthcare and education. These models leverage extensive pretraining on vast amounts of data, enabling nuanced interpretation of complex inputs and generation of empathetic, personalized feedback.

This progress opens new avenues to overcome the limitations of existing digital wellbeing tools. By integrating LLMs with passive behavioral sensing—collecting data such as mobility, ambient audio, phone usage, and academic activities without user effort—we can continuously monitor student wellbeing unobtrusively. LLMs’ deep reasoning enables interpretation of these multimodal signals to detect behavioral anomalies or risk patterns, facilitating both reactive and proactive micro-interventions tailored empathetically to each user’s context and needs.

In response to the 2025 UbiComp/ISWC Student Challenge on novel applications of LLMs in ubiquitous sensing, we present ReflectAI, a digital wellbeing prototype that combines passive sensing with LLM-based reasoning. ReflectAI processes longitudinal multimodal datasets (*StudentLife* [6] and *GLOBEM* [7]), converting raw sensor signals into semantically rich behavioral summaries (e.g., “social withdrawal,” “irregular routines”). These summaries serve as inputs for a few-shot prompting pipeline using LLMa which generates personalized, empathetic feedback such as gentle nudges, planning support, and emotional check-ins.

Unlike traditional rule-based or classifier-based systems, ReflectAI leverages LLMs’ powerful contextual and psychological reasoning to deliver context-sensitive, human-like digital wellbeing support embedded seamlessly into students’ daily lives.

Our contributions can be summarized as follows:

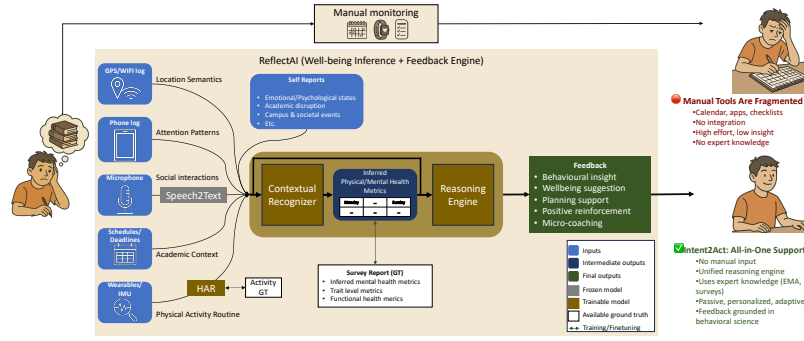


Figure 1: ReflectAI Architecture Overview integrating multimodal sensing with LLM reasoning and adaptive feedback.

- A novel system that integrates passive multimodal sensing with open-source LLMs (Deepseek R1 Distilled Qwen 7B [2], LLaMA 3B Instruct [13]) to recognize psychologically meaningful behavioral patterns and generate personalized, empathetic wellbeing feedback using few-shot prompting.
- Early design outputs with the two real-world datasets, including wireframes and storyboards illustrating the user experience.

2 Related Work

Digital Mental Health and Mobile Interventions Mobile health apps have proliferated as tools for promoting mental wellbeing, offering features such as mood tracking, cognitive behavioral therapy (CBT) exercises, and journaling [3, 12]. However, these apps typically rely on active user engagement, requiring individuals to self-report symptoms or manually initiate interaction. Studies have shown that such demands lead to low adherence, especially during times of emotional distress or academic pressure [9]. Recent reviews also highlight a lack of personalization and ecological validity in most digital interventions [10].

Large Language Models for Contextual Reasoning LLMs like GPT, PaLM, and open-source models (e.g., Qwen [2], Deepseek [8], LLaMA [13]) have recently shown strong performance in tasks requiring reasoning, interpretation, and empathy. In mental health contexts, LLMs have been applied to tasks such as therapeutic dialogue generation [11] and affective computing [14]. However, most of this work remains focused on textual input and lacks integration with real-world behavioral sensing.

ReflectAI bridges this gap by combining passive behavioral sensing with LLM-based reasoning to generate context-aware, psychologically grounded feedback. Unlike prior systems, it does not rely on manual input or rigid classifiers; instead, it leverages few-shot prompting with open-source LLMs to deliver adaptive, empathetic micro-interventions based on real-world multimodal data. This represents a novel application of LLMs in the domain of proactive, behavior-aware digital mental health.

3 System Design

ReflectAI is a proactive digital wellbeing system that transforms passive multimodal sensing data into personalized, empathetic feedback for university students. It ingests raw sensor streams and

behavioral logs, semantically interprets these signals into meaningful wellbeing states, and leverages a fine-tuned LLM reasoning model to generate context-aware micro-interventions. The system comprises three key components: data inputs and preprocessing, a contextual recognizer, an LLM reasoning and feedback engine that delivers adaptive, psychologically informed messages.

ReflectAI integrates multimodal data from the StudentLife and GLOBEM datasets, including GPS, Wi-Fi, phone usage, ambient audio, academic schedules, sleep, activity logs, and emotion annotations. GPS data are cleaned by removing outliers and interpolating missing points; daily metrics like distance traveled and time spent at key locations are aggregated over sliding windows to detect mobility changes. Wi-Fi connectivity is used to infer location stability and social proximity by tracking connections to known networks, with frequent disconnections signaling possible disengagement. Phone logs capture unlock frequency, screen time, app usage, and call/text activity, normalized daily to detect anomalies such as reduced communication or excessive use. Ambient audio features non-speech signals like ambient noise and conversation presence to approximate social interaction without storing raw audio. Academic schedules are aligned with behavioral data to flag missed classes or study sessions as indicators of disengagement or burnout. Sleep patterns are estimated from phone inactivity, accelerometer data, and self-reports to identify irregular or insufficient rest. Wearable IMU data classify physical activities to enrich mobility assessments beyond GPS metrics. These processed data streams are summarized into concise behavioral indicators (e.g., “low physical activity for 3 days,” “missed lectures,” “reduced social interaction”) that feed into contextual reasoning and LLM-based feedback generation.

Contextual Recognizer. The contextual recognizer module maps processed behavioral summaries to inferred mental health metrics, enabling a nuanced understanding of the user’s psychological state. The recognizer is trained using multimodal behavioral summaries aligned with self-reported affective states and EMA entries from the datasets. It is a fine tuned LLM capable of interpreting multimodal input and outputting psychologically meaningful, interpretable labels.

Specifically, it predicts:

- Trait-level metrics: Stable psychological traits or baseline characteristics, such as resilience or neuroticism, derived from long-term behavioral patterns.

- Functional health metrics: Dynamic, state-dependent indicators reflecting the user’s current wellbeing, including mood fluctuations, burnout levels, social withdrawal, or stress.
- Mental health status: Clinical or subclinical symptom scores, such as depression or anxiety severity, inferred using validated scales provided as ground truth in the datasets.

These inferred mental health, trait, and functional health metrics serve as enriched semantic inputs for the downstream LLM reasoning engine, facilitating personalized, context-sensitive wellbeing feedback.

LLM Reasoning Engine. ReflectAI employs pretrained LLM as its reasoning backbone. The LLM is fine-tuned on a curated dataset pairing behavioral summaries with psychologically grounded feedback, incorporating ecological momentary assessments (EMA) like PHQ-9 and PANAS scores to personalize responses.

The system tracks and stores behavioral and emotional metrics longitudinally, enabling temporal context awareness. By leveraging the LLM’s pretrained reasoning capabilities, ReflectAI detects anomalies or deviations from typical patterns over time—such as sudden drops in activity or mood—and generates constructive, context-sensitive feedback aimed at improving mental wellbeing.

Training data for fine-tuning are constructed by converting behavioral summaries and emotional baselines into natural language prompts, paired with target feedback texts generated by clinical experts or derived from psychology literature. During inference, few-shot prompting is used to produce empathetic, adaptive micro-interventions that are sensitive to both current and historical user states.

The feedback engine operationalizes LLM outputs into user-facing interventions, delivering personalized micro-coaching messages and adaptive check-ins timed to the user’s behavioral rhythms and receptiveness. It applies affective computing principles to modulate tone and emotional supportiveness, reducing user fatigue and enhancing engagement over time.

Training Objective and Loss Function. ReflectAI involves one training and two fine-tuning stages with distinct objectives:

HAR Model Training: The HAR model is trained using a cross-entropy loss that minimizes the discrepancy between predicted activity classes and ground-truth labels from the IMU data. The loss is defined as:

$$\mathcal{L}_{HAR} = - \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log \hat{y}_{i,c} \quad (1)$$

where N is the number of samples, C is the number of activity classes, $y_{i,c}$ is the true label indicator (1 if sample i belongs to class c , otherwise 0), and $\hat{y}_{i,c}$ is the predicted probability for class c .

LLM Fine-tuning: The llms used in both the contextual recognizer and reasoning engine are fine-tuned on paired datasets of behavioral summaries and expert-curated feedback. The training optimizes the cross-entropy loss over token sequences:

$$\mathcal{L}(\theta) = - \sum_{t=1}^m \log P_{\theta}(y_t | y_{<t}, X) \quad (2)$$

where X is the input prompt sequence representing behavioral context, and Y is the target feedback text. This fine-tuning enables the LLMs to generate empathetic, context-aware interventions grounded in psychological theory.

4 Initial Results

4.1 Quantitative Evaluation

We evaluated ReflectAI’s performance in generating context-aware wellbeing feedback using multimodal data from the StudentLife and GLOBEM datasets over Deepseek R1 Distilled Qwen 7B [2] and LLaMA 3B Instruct [13] backbone. Both datasets include ground-truth psychological assessments scores as well as manually derived , enabling quantitative validation of the system’s inferred mental health metrics. To assess the ReflectAI’s ability to generate psychologically relevant and personalized feedback, we computed correlations between ReflectAI’s inferred mental health scores and the corresponding self-reported survey measures for both datasets. Table 1 summarizes these results.

Table 1: Correlation between ReflectAI-inferred mental health metrics and ground-truth assessments across models and datasets

Metric	Model	Dataset	Correlation (r)	p-value
PHQ-9	Qwen 7B	StudentLife	0.66	0.004
	LLaMA 3 8B	StudentLife	0.70	0.002
	Qwen 7B	GLOBEM	0.61	0.007
	LLaMA 3 8B	GLOBEM	0.65	0.004
PANAS (+)	Qwen 7B	StudentLife	0.60	0.010
	LLaMA 3 8B	StudentLife	0.64	0.006
	Qwen 7B	GLOBEM	0.56	0.021
	LLaMA 3 8B	GLOBEM	0.60	0.012
PANAS (-)	Qwen 7B	StudentLife	0.58	0.015
	LLaMA 3 8B	StudentLife	0.62	0.008
	Qwen 7B	GLOBEM	0.53	0.032
	LLaMA 3 8B	GLOBEM	0.58	0.018

These moderate to strong correlations indicate that ReflectAI effectively captures psychological states from multimodal behavioral summaries, despite relying on passive sensing data without direct symptom reporting.

4.2 Qualitative Feedback Examples

ReflectAI generates empathetic, psychologically grounded micro-interventions by synthesizing behavioral patterns from multimodal sensor data and contextual information. Below are representative feedback examples that align with specific behavioral anomalies detected in the StudentLife and GLOBEM datasets:

- Low mobility + 3 missed morning classes: “It looks like you’ve been staying indoors and missing your morning lectures. Would a gentle routine or a morning walk help you reset the day?”
- Reduced phone communication + no Wi-Fi switches : “You haven’t been as socially connected lately. Want to check in with a friend or explore a quiet campus spot for a change of scene?”

- Late sleep and irregular activity patterns: “Your recent sleep times have been quite late. Establishing a wind-down ritual might help your body and mind recharge better.”

These examples illustrate ReflectAI’s capacity to convert passive sensing data into meaningful, non-intrusive, and psychologically supportive interventions tailored to students’ real-life patterns.

While promising, these results remain preliminary and based on retrospective dataset analyses. Comprehensive real-time deployment and longitudinal user studies are essential to rigorously validate ReflectAI’s effectiveness in practical, everyday settings. Such studies would also help identify potential usability challenges, optimize feedback timing and content, and refine personalization strategies to better accommodate diverse student needs and preferences. Moreover, assessing the system’s impact on mental health outcomes over time will be crucial for establishing its clinical and educational value.

5 Ethical Implications

User Benefit. ReflectAI is designed to support students through passive sensing and LLM-based reflection, reducing the burden of manual self-tracking. It aims to increase awareness of behavioral trends (e.g., isolation, disrupted routines) and offer empathetic, context-sensitive feedback that encourages proactive coping strategies. ReflectAI may be especially valuable for students who are under-supported or reluctant to seek formal mental health services. By intervening early, it could help mitigate stress, burnout, and disengagement without requiring therapy or diagnosis. Over time, such feedback may also help users build reflective habits that promote long-term resilience and emotional self-regulation.

Beyond Privacy: Potential Misuse. While privacy concerns are well-studied in sensor-driven systems, ReflectAI presents additional risks stemming from its interpretive and generative capabilities: If deployed institutionally, ReflectAI could be repurposed for performance monitoring, behavioral scoring, or disciplinary action. LLM-generated messages, even when empathetic, may exert subtle influence over user behavior, nudging them toward perceived “healthy” norms that may not be contextually appropriate. Inference errors may lead to inappropriate or inaccurate feedback, which could worsen distress, trigger anxiety, or erode user trust. Despite careful fine-tuning, large language models may embed cultural, psychological, or normative biases that result in insensitive or misaligned feedback across diverse user populations.

Mitigation Strategies. To reduce these risks, ReflectAI incorporates multiple safeguards by design. All feedback is delivered as optional suggestions, never as prescriptions. Users can opt out, adjust feedback tone, and access system reasoning (e.g., “Why am I seeing this?”). Feedback is accompanied by plain-language rationales derived from behavioral summaries (e.g., “Based on reduced movement and missed events...”), supporting interpretability. System defaults prioritize mental wellbeing over productivity, avoiding over-optimization pressures or competitive comparisons. Users are transparently informed of what data is collected, how it is used, and when processing occurs on-device versus server-side.

Long-Term Social Considerations. ReflectAI, like other passive monitoring systems, may contribute to shifting norms around self-surveillance, behavioral nudging, and digital autonomy. While intended to support user agency, repeated interactions with AI-generated reflections may affect how individuals understand and regulate their own behavior. To mitigate these effects, we explicitly frame ReflectAI as a research prototype, not a clinical or institutional deployment tool. Future work will explore its impact on mental wellbeing, sense of agency, and user autonomy in real-world contexts, with a strong emphasis on user-centered co-design and participatory governance.

6 Conclusion

ReflectAI leverages the synergy of multimodal sensing and LLM reasoning to provide proactive, personalized wellbeing support tailored to university students. Its adaptive feedback mechanism offers a promising alternative to manual self-monitoring and static rule-based systems, with potential applications in digital mental health interventions, education, and personal AI companions. Future work will focus on user studies to validate efficacy and refine engagement strategies.

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